



Nebraska Public Power District

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10 CFR 50.73

NLS2023001
January 11, 2023

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2022-003-00
Cooper Nuclear Station, Docket No. 50-298, DPR-46

Dear Sir or Madam:

The purpose of this correspondence is to forward Licensee Event Report 2022-003-00.

There are no regulatory commitments contained in this letter.

Sincerely,

Khalil Dia
Site Vice President

/jo

Attachment: Licensee Event Report 2022-003-00

cc: Regional Administrator w/attachment
USNRC - Region IV

NPG Distribution w/attachment

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

INPO Records Center w/attachment
via IRIS entry

Senior Resident Inspector w/attachment
USNRC - CNS

SORC Chairman w/attachment

SRAB Administrator w/attachment

CNS Records w/attachment

NRC FORM 366 (08-2020)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104				EXPIRES: 08/31/2023			
		LICENSEE EVENT REPORT (LER) (See Page 3 for required number of digits/characters for each block)			(See NUREG-1022, R.3 for instruction and guidance for completing this form https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)						Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov . The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.	
1. Facility Name Cooper Nuclear Station					2. Docket Number 05000 298			3. Page 1 OF 4				
4. Title Actuation of Reactor Protection System Initiates a Full Reactor Scram												
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved			
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number		
11	12	2022	2022	- 003	00	01	11	2023	Facility Name	05000		
9. Operating Mode					10. Power Level							
2					002							
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)												
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)				
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73				
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(4)				
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.71(a)(5)				
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(1)(i)				
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(i)				
☐ 20.2203(a)(2)(iii)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.77(a)(2)(ii)				
☐ 20.2203(a)(2)(iv)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)						
☐ 20.2203(a)(2)(v)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)						
☐ Other (Specify here, in Abstract, or in NRC 366A).												
12. Licensee Contact for this LER												
Licensee Contact Linda Dewhirst, Regulatory Affairs and Compliance Manager								Phone Number (Include Area Code) (402) 825-5416				
13. Complete One Line for each Component Failure Described in this Report												
Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS			
				Y								
14. Supplemental Report Expected								15. Expected Submission Date				
☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)								Month Day Year				
16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)												
On November 12, 2022, during plant start up from Refueling Outage 32, operations personnel commenced a surveillance for High Pressure Coolant Injection (HPCI) Beginning of Cycle Testing. While performing the test, a HPCI injection valve unexpectedly opened, injecting water into the reactor vessel. The injected water increased reactivity and the Intermediate Range Monitors initiated a reactor scram. As such, the test was stopped. In April 2021, an undervoltage relay for the valve had been removed due to reliability concerns with that specific type of relay – if the relay would fail, it would cause the valve to fail. Investigation revealed that the modification process to remove the undervoltage relay did not identify a change was required to the HPCI surveillance procedure. There was no impact on nuclear safety, plant reliability, radiological safety, or industrial safety. This event was reported under Event Notification 56220 on November 12, 2022.												

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Cooper Nuclear Station	05000- 298	YEAR	SEQUENTIAL NUMBER	REV NO.
		2022	- 003	- 00

NARRATIVE**PLANT STATUS**

Cooper Nuclear Station was in Mode 2, Startup, at 2 percent power at the time of the event on November 12, 2022.

BACKGROUND

The Reactor Protection System (RPS) provides timely protection against the onset and consequences of conditions that threaten the integrity of the fuel barrier and the reactor coolant pressure boundary. The RPS includes sensors, relays [EIS: RLY], bypass circuits, and switches that are necessary to cause initiation of a reactor [EIS: RPV] scram.

The Intermediate Range Monitors (IRM) generate a trip signal that can be used to prevent fuel damage resulting from abnormal operational transients that occur while operating in the intermediate power range.

The Emergency Core Cooling System (ECCS) [EIS: BJ, BM, BO] control and instrumentation initiates appropriate response from the various cooling systems so fuel is adequately cooled under abnormal and accident conditions. The controls and instrumentation for the ECCS are identified as that equipment required for the initiation and control of the High Pressure Coolant Injection System (HPCI), the Automatic Depressurization System, the Core Spray System, and the Low Pressure Coolant Injection System.

When actuated, the HPCI system pumps water from either the emergency condensate [EIS: SD] storage tanks (ECST) or the suppression chamber to the reactor vessel via the feedwater pipeline. 250 volt direct current (VDC) provides motive power and 125 VDC provides control power for pumps [EIS: P] and valves [EIS: V].

At the beginning of the operating cycle, HPCI is tested to assess operational readiness prior to exceeding 165 pounds per square inch gauge. The surveillance aligns the system in the test return flow lineup, by placing HPCI-MO-21 (test bypass valve to ECST) and HPCI-MO-24 (ECST test line shutoff valve) in the OPEN position, and HPCI-MO-19 (injection valve) in the CLOSED position. 250 VDC power to the three valves is removed to prevent operation of the valves during the surveillance. A test switch is used to initiate the HPCI response. The HPCI system logic provides a CLOSE signal to HPCI-MO-21 and HPCI-MO-24, and an OPEN signal to HPCI-MO-19 via relay contacts. After HPCI system flow is verified and HPCI component functions are verified to have operated correctly, the test switch is placed to OFF, de-energizing the relays and the HPCI initiation signal is reset de-energizing other relays. 250 VDC power to the HPCI-MO-21, HPCI-MO-24, and HPCI-MO-19 valves is then restored.

EVENT DESCRIPTION

On November 12, 2022, during plant start up from Refueling Outage 32 (RE32), operations personnel commenced surveillance procedure HPCI Beginning of Cycle Test. The purpose of this surveillance is to

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perform a HPCI system simulated automatic actuation test and assess the HPCI system operational readiness.

At 22:40, HPCI was declared inoperable to perform the surveillance test. While performing the surveillance, following restoration of 250 VDC power to HPCI-MO-21, HPCI-MO-24, and HPCI-MO-19 valves, at 23:18 HPCI-MO-19 unexpectedly opened and water was injected into the reactor vessel via the "B" feedwater line. The injected water increased reactivity and the IRMs initiated a reactor scram at 23:19. As a result, the surveillance was stopped. HPCI-MO-19 was subsequently closed and on November 13, at 01:26 HPCI was restored in standby and OPERABLE status.

Subsequent to the event, it was discovered that an undervoltage relay monitoring the 250 VDC power for HPCI-MO-19 had been removed in April 2021 by a design modification, due to reliability concerns with that specific type of relay – if the relay would fail, it would cause the valve to fail. The undervoltage relay had a contact in the 125 VDC control circuit which prevented an OPEN signal from "sealing in" for the HPCI-MO-19 valve if the 125 VDC control power was energized when the 250 VDC power was removed. With the undervoltage relay and its contact removed, the test switch actuated the OPEN logic which remained active when the 250 VDC power was restored.

Investigation revealed that the modification process to remove the undervoltage relay did not identify a change was required to the HPCI surveillance procedure.

BASIS FOR REPORT

This event is reportable under 10 CFR 50.73(a)(2)(iv)(A) – Any event or condition that resulted in manual or automatic actuation of any of the systems listed in 10 CFR 50.73(a)(2)(iv)(B); specifically, (a)(2)(iv)(B)(1) – Reactor protection system (RPS) including: reactor scram or reactor trip; and (a)(2)(iv)(B)(4) – ECCS for boiling water reactors including high-pressure and low-pressure core spray systems, high-pressure coolant injection system, and low-pressure injection function of the residual heat removal system.

This event was also reported under Event Notification 56220 on November 12, 2022.

SAFETY SIGNIFICANCE

This event is non-consequential. The actual consequences were related to a reactor power increase resulting in an automatic neutron flux scram. The scram occurred with no complications. Investigation indicated that mitigating systems required for the mode of operation were available and responded as designed. There were no equipment failures that contributed to this event.

There was no impact to nuclear safety, plant reliability, radiological safety, or industrial safety.

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CAUSE

The direct cause of this event was the removal of an undervoltage relay and its contact, such that when the test switch actuated the OPEN logic for HPCI-MO-19, the signal remained active, resulting in the valve to unexpectedly open when 250 VDC power was restored.

There were two causal factors identified. The first causal factor was the modification process did not require use of the procedure for design inputs or other tools to require descriptions of operational impacts; specifically for surveillance equipment lineups. The second causal factor was potential impacts from engineering change documents to surveillance lineups were not well understood.

CORRECTIVE ACTIONS

The surveillance procedure was revised to remove the 125 VDC control power for both HPCI-MO-21 and HPCI-MO-19 after opening the associated 250 VDC breaker. Restoration of 250 VDC power for HPCI-MO-19 was moved to after the HPCI turbine is tripped and pump discharge valve is closed.

The modification preparation, review and approval process will be revised.

PREVIOUS EVENTS

On October 22, 2022, Cooper Nuclear Station (CNS) control room operators manually injected core spray to restore skimmer surge tank level. This was reported under LER 2022-002-00, Manual Core Spray Injection to Restore Skimmer Surge Tank Level.